

Dynamical Calibration of a Metallicity-Dependent Mass–Luminosity Relation from Gaia Wide Binaries

Frozen model configuration: 13 September 2026

1 Scientific objective and modular architecture

The aim is to infer the stellar mass M as a function of Gaia absolute G -band magnitude $x \equiv M_G$ and metallicity $Z \equiv [\text{M}/\text{H}]$, while retaining the astrophysical mass scale supplied by the PARSEC stellar-evolution models (Bressan et al., 2012). Wide binaries are useful because their components are assumed to share a birth composition, whereas their relative motion constrains the sum of their masses. The data therefore connect two otherwise separate calibration problems: the metallicity assigned to each system and the mass–luminosity relation (MLR).

The Gaia astrometry and broad-band photometry used here belong to the third Gaia data-release framework (Gaia Collaboration et al., 2023); the relevant G , G_{BP} , and G_{RP} photometric processing and validation are described by Riello et al. (2021). Large Gaia wide-binary catalogues and their principal selection and astrometric-validation issues are described by El-Badry et al. (2021).

We use modular inference (Liu et al., 2009; Plummer, 2015). Stage 1a uses within-binary metallicity differences to estimate the relative JCAPS calibration. Conditional on that estimate, Stage 1b combines the component JCAPS measurements with the PARSEC colour–magnitude diagram (CMD) and a hierarchical population distribution to infer a probability distribution for the shared metallicity of every binary. Stage 2 treats these distributions as fixed inputs, predicts the two component masses at every metallicity, and evaluates the wide-binary dynamical likelihood. The modular inference deliberately prevents the dynamical data from recalibrating the JCAPS zero point, CMD scatter, or sample metallicity distribution. The architecture is summarized in figure 1.

2 Observables, latent quantities, and conventions

Binary systems are indexed by j , their two components by $k \in \{1, 2\}$, and metallicity nodes by q . A hat denotes an observed quantity. The symbol $\log_{10}$ is reserved for base-10 stellar-mass logarithms; unqualified $\log$ denotes the natural logarithm. “Dex” denotes an interval in a base-10 logarithm. The main quantities are listed in table 1.

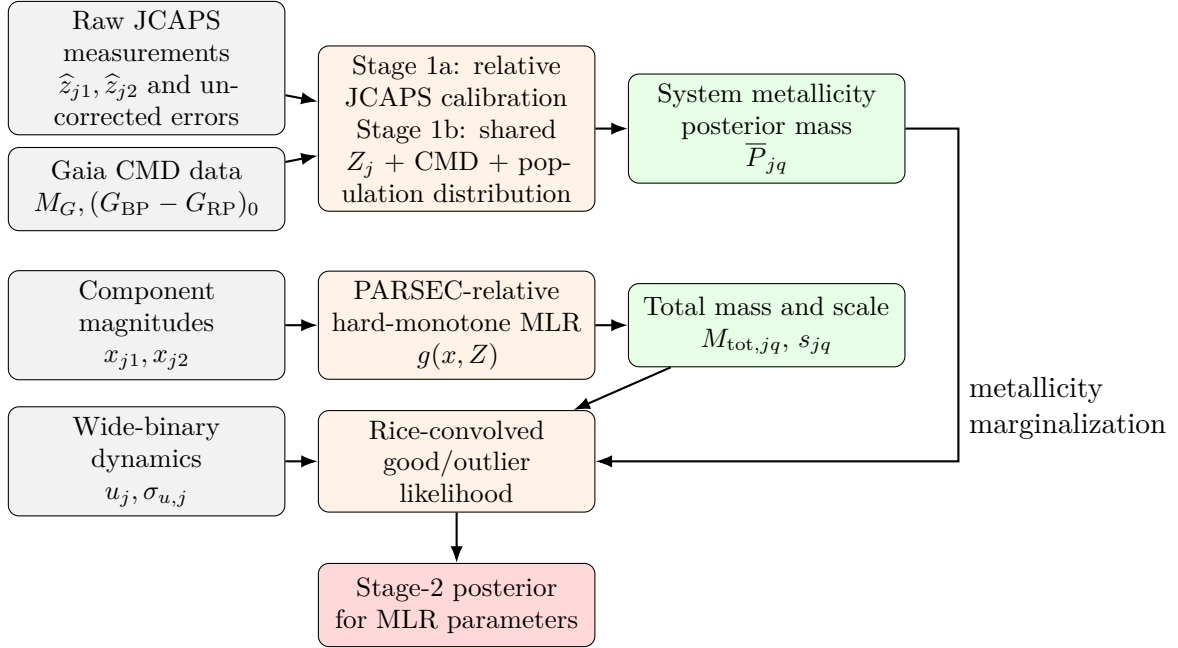


Figure 1: Scientific workflow. Stage 1a estimates the relative JCAPS calibration from within-binary differences. Conditional on those fixed estimates, Stage 1b returns a posterior probability mass over each shared system metallicity. Stage 2 propagates this uncertainty into the dynamical MLR fit but does not update either Stage 1 module. This one-way flow implements the modular cut.

Table 1: Principal observables and latent or derived quantities.

Symbol	Meaning	Unit or range
$\mathcal{D}_{1a}$	Within-binary JCAPS differences used to estimate θ_{rel}	data set
$\mathcal{D}_{1b}$	JCAPS and Gaia CMD data used for absolute calibration and system posteriors	data set
$\mathcal{D}_2$	Stage-2 component-magnitude and wide-binary dynamical data, with $\bar{P}_{jq}$ supplied from Stage 1b	data set
x_{jk}	Gaia absolute magnitude of component k	mag
C'_{jk}	Dereddened colour $(G_{\text{BP}} - G_{\text{RP}})_0$	mag
$\hat{z}_{jk}$	Raw JCAPS XP metallicity measurement	dex
$\sigma_{z,jk}$	Reported, uncorrected JCAPS uncertainty	dex
Z_j	Shared true system metallicity on the PARSEC scale	dex
$\bar{P}_{jq}$	Stage-1b posterior probability mass at Z_q	[0, 1]
$g(x, Z)$	Final $\log_{10}(M/M_{\odot})$ surface	dex
u_j	Observed $\Delta v_{\text{tan}} \sqrt{r_{\perp}}$ amplitude	$\text{km s}^{-1} \sqrt{\text{AU}}$
$\sigma_{u,j}$	Component-wise Gaussian error scale entering the Rice model	$\text{km s}^{-1} \sqrt{\text{AU}}$
s_{jq}	Dynamical mass scale $\sqrt{M_{\text{tot},jq}/M_{\odot}}$	dimensionless

The metallicity inputs are strictly `feh_jcaps_1`, `feh_jcaps_2`, `jc_sigma_m_h_1`, and `jc_sigma_m_h_2`. JCAPS infers stellar labels by inverting a forward model of Gaia XP spectra; the underlying framework and its training strategy are described by Li et al. (2026). The names above refer to columns in the analysis table and are not asserted to be published JCAPS catalogue field names. No binary-corrected, fitted, calibrated, or inflated metallicity column is used. Such columns already incorporate the equal-composition assumption that appears in our likelihood; reusing them would count that information twice and make the calibration circular.

3 Stage 1: relative and absolute metallicity calibration

3.1 Stage 1a: Relative JCAPS calibration

Before introducing the PARSEC CMD, the relative behavior of JCAPS is calibrated from the difference between the two stars in each binary. The adopted observation model assigns one independent Student- t state to each star,

$$\hat{z}_{jk} \mid Z_j, \delta_z \sim \mathcal{T}_\nu \left[Z_j + \delta_z + b_G(x_{jk}), \sqrt{\sigma_{z,jk}^2 + s_z^2} \right] = \mathcal{L}_{jk}^{\text{spec}}(Z_j), \quad (1)$$

where the second argument is the location and the third is the scale. The function $b_G(x)$ is a magnitude-dependent *relative* bias, s_z is one common additional scale per star (hence a contribution $2s_z^2$ to a binary difference variance), and ν is the shared degrees of freedom. Conditional on Z_j , the two component measurements are independent. No quality-dependent mixture state, χ^2 -dependent anomaly probability, catastrophic -1.5 -dex component, or shared poor-observation offset is included.

More explicitly, let $d_j = \hat{z}_{j1} - \hat{z}_{j2}$, $\Delta b_j = b_G(x_{j1}) - b_G(x_{j2})$, and $a_{jk} = (\sigma_{z,jk}^2 + s_z^2)^{1/2}$. The within-binary difference likelihood is the convolution

$$p(d_j \mid \Delta b_j, s_z, \nu) = \int_{-\infty}^{\infty} \mathcal{T}_\nu(e \mid 0, a_{j1}) \mathcal{T}_\nu(e - [d_j - \Delta b_j] \mid 0, a_{j2}) de. \quad (2)$$

The convolution of two Student- t variables is not, in general, another Student- t distribution. Both Z_j and the common offset δ_z cancel from d_j , but the observed differences still constrain the relative-bias shape b_G , the per-star excess scale s_z , and the common tail parameter ν . Stage 1a obtains the maximum-likelihood estimate

$$\hat{\theta}_{\text{rel}} = \arg \max_{\theta_{\text{rel}}} \sum_{j \in \mathcal{D}_{1a}} \log p(d_j \mid \theta_{\text{rel}}), \quad \theta_{\text{rel}} = (b_G, s_z, \nu), \quad (3)$$

and checks the resulting difference distribution on a held-out subset that is not used in this maximization. The differences cannot determine an absolute metallicity zero point. We set $b_G(8.5) = 0$ and use the frozen plug-in values

$$x_{\text{knot}}/\text{mag} = (3.5, 6.0, 8.5, 11.0, 13.5), \quad (4)$$

$$b_G(x_{\text{knot}})/\text{dex} = (0.11, 0.05, 0, -0.31, -0.53), \quad (5)$$

$$s_z = 0.05 \text{ dex}, \quad \nu = 2.5, \quad (6)$$

with linear interpolation between knots. Stage 1b conditions on $\hat{\theta}_{\text{rel}}$: these three quantities are fixed inputs, not parameters assigned priors or re-estimated in Stage 1b.

3.2 Stage 1b: Absolute calibration and system metallicity posteriors

3.2.1 Hierarchical population distribution

The latent metallicities of the selected systems follow

$$p_{\text{pop}}(Z \mid \omega) = \sum_{r=1}^6 \omega_r \varphi_r^{\text{trunc}}(Z), \quad \omega \sim \text{Dirichlet}(1, \dots, 1), \quad (7)$$

where every φ_r^{trunc} is a Gaussian density renormalized on $-1.0 \leq Z \leq 0.6$ dex. Writing ϕ and Φ for the standard-normal density and cumulative distribution function, respectively,

$$\varphi_r^{\text{trunc}}(Z) = \frac{\phi[(Z - \mu_r)/\tau_r]}{\tau_r \{ \Phi[(Z_{\text{max}} - \mu_r)/\tau_r] - \Phi[(Z_{\text{min}} - \mu_r)/\tau_r] \}} \mathbb{I}(Z_{\text{min}} \leq Z \leq Z_{\text{max}}). \quad (8)$$

Here $Z_{\text{min}} = -1.0$ and $Z_{\text{max}} = 0.6$ dex, the six centers μ_r are equally spaced between these limits, and every $\tau_r = 1.6/6 = 0.2667$ dex. This hierarchical population distribution is learned jointly with the other Stage-1b parameters; it is not an independently estimated external prior. Because it describes the systems after sample selection, it must not be interpreted as the intrinsic Galactic metallicity distribution function.

The remaining common offset is assigned

$$\delta_z \sim \mathcal{N}(0, 0.3^2) \quad \text{dex}. \quad (9)$$

Under the sign convention of [equation \(1\)](#), positive δ_z means that JCAPS measurements lie above the PARSEC metallicity scale at fixed Z_j .

3.2.2 PARSEC CMD likelihood and the 4000-K mask

The PARSEC surface supplies predicted colour $C_P(x, Z)$ and effective temperature $T_{\text{eff},P}(x, Z)$ ([Bressan et al., 2012](#)). The repository records the age, metallicity, and magnitude coverage of the local PARSEC table, but not enough provenance to identify a more specific PARSEC release or Gaia passband realization; we therefore make no more specific version claim. The formal colour uncertainty is propagated from the BP and RP signal-to-noise ratios,

$$\sigma_{C,jk} = \frac{2.5}{\ln 10} \left(\text{SNR}_{\text{BP},jk}^{-2} + \text{SNR}_{\text{RP},jk}^{-2} \right)^{1/2}. \quad (10)$$

For an eligible component, the CMD likelihood is

$$C_{jk} \mid Z_j \sim \mathcal{T}_4 \left[C_P(x_{jk}, Z_j), \sqrt{\sigma_{C,jk}^2 + s_C^2} \right] = \mathcal{L}_{jk}^{\text{CMD}}(Z_j). \quad (11)$$

The colour zero point is fixed to zero and $s_C \sim \text{HalfNormal}(0.1 \text{ mag})$. To avoid allowing membership in the CMD term to change with a proposed metallicity, eligibility is fixed over the full latent-metallicity domain:

$$m_{jk} = \mathbb{I} \left[\min_{q \in \mathcal{Q}} T_{\text{eff},P}(x_{jk}, Z_q) \geq 4000 \text{ K} \right]. \quad (12)$$

If $m_{jk} = 0$, that star still contributes its JCAPS likelihood but not its CMD likelihood. To construct the fixed CMD reference surface, we selected PARSEC rows with $9.3 < \log_{10}(\text{age/yr}) < 10.0$ and $3 < G_{\text{mag}} < 15$. At fixed source-table G_{mag} and metallicity, rows from the selected ages were averaged separately in colour and $\log T_{\text{eff}}$, so age is not a latent variable in this analysis. Colour was represented by an up-to-cubic smoothing spline with smoothing parameter 0.001; $\log T_{\text{eff}}$ was linearly interpolated and then exponentiated before tabulation. The colour and linear- T_{eff} tables were evaluated bilinearly in (M_G, Z) on a 500-point magnitude grid spanning $3.5 \leq M_G \leq 13.5$. This CMD construction is distinct from the 501×81 (M_G, Z) grid used only to fit the MLR's fixed PARSEC mass projection.

The frozen sample contains 14,876 binaries (29,752 stars). The 4000-K rule admits 12,167 stars; 8,247 systems have at least one eligible component and 3,920 have two. These counts describe CMD coverage, not evidence that the cool-star PARSEC colours are correct.

3.2.3 System posterior and the cut object

The sampled Stage-1b global parameters are exactly $\boldsymbol{\psi} = (\delta_z, s_C, \omega)$. Conditional on $\hat{\boldsymbol{\theta}}_{\text{rel}}$, the unnormalized metallicity density for one system is

$$\tilde{p}_j(Z | \boldsymbol{\psi}, \hat{\boldsymbol{\theta}}_{\text{rel}}) = p_{\text{pop}}(Z | \omega) \prod_{k=1}^2 \mathcal{L}_{jk}^{\text{spec}}(Z | \delta_z, \hat{\boldsymbol{\theta}}_{\text{rel}}) \left[\mathcal{L}_{jk}^{\text{CMD}}(Z | s_C) \right]^{m_{jk}}. \quad (13)$$

The global parameters are inferred from the product of the system-level marginal evidences, rather than from separately normalized system posteriors:

$$p(\boldsymbol{\psi} | \mathcal{D}_{1b}, \hat{\boldsymbol{\theta}}_{\text{rel}}) \propto p(\delta_z) p(s_C) p(\omega) \prod_j \left[\sum_{q \in \mathcal{Q}} w_q^{(Z)} \tilde{p}_j(Z_q | \boldsymbol{\psi}, \hat{\boldsymbol{\theta}}_{\text{rel}}) \right]. \quad (14)$$

The model evaluates this expression at 81 equally spaced nodes $Z_q \in [-1.0, 0.6]$ dex using trapezoidal integration weights $w_q^{(Z)}$:

$$P_{jq}(\boldsymbol{\psi}, \hat{\boldsymbol{\theta}}_{\text{rel}}) = \frac{w_q^{(Z)} \tilde{p}_j(Z_q | \boldsymbol{\psi}, \hat{\boldsymbol{\theta}}_{\text{rel}})}{\sum_{q'} w_{q'}^{(Z)} \tilde{p}_j(Z_{q'} | \boldsymbol{\psi}, \hat{\boldsymbol{\theta}}_{\text{rel}})}. \quad (15)$$

The saved quantity passed to Stage 2 is the posterior average

$$\bar{P}_{jq} = \mathbb{E}_{\boldsymbol{\psi} | \mathcal{D}_{1b}, \hat{\boldsymbol{\theta}}_{\text{rel}}} \left[P_{jq}(\boldsymbol{\psi}, \hat{\boldsymbol{\theta}}_{\text{rel}}) \right]. \quad (16)$$

Each system helps learn the shared hyperparameters $\boldsymbol{\psi}$ and is then partially pooled through their hyperposterior. This standard hierarchical self-influence is not likelihood double-counting: each observation enters the joint Stage-1b likelihood once. The average in [equation \(16\)](#) integrates over the global hyperposterior rather than plugging in a single value of $\boldsymbol{\psi}$. Thus Stage 2 receives a probability distribution rather than a metallicity point estimate. Because $\bar{P}_{jq}$ already includes p_{pop} , both spectroscopic likelihoods, every eligible CMD likelihood, and the metallicity quadrature weights, Stage 2 must not multiply any of these population or quadrature factors again.

4 Stage 2: a PARSEC-relative hard-monotone MLR

4.1 Baseline projection and scientific correction

Let the original PARSEC mass surface be

$$g_{\text{P}}(x, Z) = \log_{10} \left[\frac{M_{\text{P}}(x, Z)}{M_{\odot}} \right]. \quad (17)$$

The finite MLR basis cannot reproduce this tabulated surface exactly. The projected PARSEC mass surface onto the same tensor-product B-spline basis used by the fitted model is represented as $g_{\text{P}}^*(x, Z)$,

$$g_{\text{P}}^*(x, Z) = \sum_{i=1}^{K_x} \sum_{h=1}^{K_Z} \Pi_{ih} B_i(x) C_h(Z), \quad (18)$$

where Π is fixed. The fitted correction is

$$\Delta_*(x, Z) = \sum_{i,h} D_{ih} B_i(x) C_h(Z), \quad \Theta_{ih} = \Pi_{ih} + D_{ih}, \quad (19)$$

and hence the fitted mass surface is

$$g(x, Z) = \sum_{i,h} \Theta_{ih} B_i(x) C_h(Z) = g_{\text{P}}^*(x, Z) + \Delta_*(x, Z). \quad (20)$$

The quantity reported scientifically relative to the original PARSEC table is not Δ_* alone but

$$\Delta_P(x, Z) = g(x, Z) - g_P(x, Z) = \Delta_*(x, Z) + \epsilon_P(x, Z), \quad \epsilon_P = g_P^* - g_P. \quad (21)$$

where ϵ_P is the deterministic projection error, the RMS and maximum absolute values of ϵ_P are 0.002 and 0.019 dex, respectively.

The adopted basis has $K_x \times K_Z = 8 \times 4$ cubic B-splines (de Boor, 2001). Both knot vectors are open and clamped:

$$\kappa_x = (3.5, 3.5, 3.5, 3.5, 5.5, 7.5, 9.5, 11.5, 13.5, 13.5, 13.5, 13.5) \text{ mag}, \quad (22)$$

$$\kappa_Z = (-1.0, -1.0, -1.0, -1.0, 0.6, 0.6, 0.6, 0.6) \text{ dex}. \quad (23)$$

4.2 Hard monotonicity of the total mass surface

At fixed metallicity, increasing M_G means a fainter star and must not increase its mass. At fixed magnitude, the adopted physical assumption is that increasing metallicity must not decrease the mass. We therefore require

$$\Theta_{i+1,h} \leq \Theta_{ih}, \quad \Theta_{i,h+1} \geq \Theta_{ih}. \quad (24)$$

Because B-spline derivatives are non-negative lower-degree basis functions weighted by adjacent control-coefficient differences, these inequalities imply throughout the continuous domain

$$\frac{\partial g}{\partial x} \leq 0, \quad \frac{\partial g}{\partial Z} \geq 0. \quad (25)$$

The constraints apply to the *total* coefficient surface $\Theta = \Pi + D$, not to the correction D alone. The correction is consequently free to increase or decrease in either coordinate. We also impose no sign constraint on its mixed differences; doing so would add an unsupported assumption about how the magnitude slope changes with metallicity. The constructive parameterization that enforces equation (24) is given in appendix A.

5 Wide-binary dynamical likelihood

5.1 Mass-scaled velocity statistic

In binary orbital motion, the system mass ($m_{\text{tot}} = m_1 + m_2$), component separation (a), true anomaly θ and eccentricity e set the instantaneous relative speed,

$$v_{\text{rel}} = \sqrt{\frac{G(m_1 + m_2)}{a}} f(\theta, e), \quad (26)$$

with $f(\theta, e)$ encapsulating the phase dependence along the ellipse. At pericenter ($\theta = 0$),

$$v_{\text{rel,peri}} = \sqrt{\frac{G(m_1 + m_2)}{a}} \sqrt{\frac{1+e}{1-e}}, \quad (27)$$

and at apocenter ($\theta = \pi$),

$$v_{\text{rel,apo}} = \sqrt{\frac{G(m_1 + m_2)}{a}} \sqrt{\frac{1-e}{1+e}}. \quad (28)$$

To connect physics with observables we define

$$u := v\sqrt{s}, \quad (29)$$

where v is the projected relative speed and s is the projected separation. Under this definition, $u \propto \sqrt{m_{\text{tot}}/M_\odot}$ and depends on nuisance parameters θ , e , and orientation ϕ . We again separate

the mass dependence by defining $\tilde{u} \equiv u/\sqrt{m_{\text{tot}}/M_{\odot}}$, so $\tilde{u}$ depends only on the nuisance parameters and their influence can be marginalized. We perform simulations and pre-fit $p(\tilde{u})$ by integrating $p(\tilde{u}|\phi, \theta, e)$ over the nuisance distributions. For an observed u the likelihood then becomes

$$p(u|m_{\text{tot}}) = \frac{1}{\sqrt{m_{\text{tot}}/M_{\odot}}} p\left(\tilde{u} \equiv \frac{u}{\sqrt{m_{\text{tot}}/M_{\odot}}}\right). \quad (30)$$

Our goal is to constrain the mass–luminosity mapping $\mathcal{M}$, which links photometric observables to stellar mass. We treat

$$\{M_{\text{G}}, \text{BP} - \text{RP}, Z\} = \mathcal{M}(m, Z), \quad (31)$$

with parameters Θ defining the functional form. Conversely, stellar masses are obtained from the inverse map $m = \mathcal{M}^{-1}(M_{\text{G}}, Z; \Theta)$. The likelihood for one measurement is then

$$\begin{aligned} \mathcal{L}_j & \\ &= p\left(u_j | m_{\text{tot}}; \mathcal{M}^{-1}(M_{\text{G},j1}, Z_j; \Theta) + \mathcal{M}^{-1}(M_{\text{G},j2}, Z_j; \Theta)\right). \end{aligned} \quad (32)$$

Measurement uncertainties on the tangential velocity enter mainly through the uncertainty of the proper-motion difference $\Delta\mu$ in the relation of

$$u = 4.74 \times \frac{\Delta\mu}{\varpi} \times \sqrt{s}. \quad (33)$$

Assuming Rice-distributed proper-motion differences,

$$\begin{aligned} \text{Rice}(\Delta\mu_{\text{obs}} | \Delta\mu_{\text{true}}, \sigma_{\Delta\mu}) &= \frac{\Delta\mu_{\text{obs}}}{\sigma_{\Delta\mu}^2} \\ &\times \exp\left[-\frac{\Delta\mu_{\text{obs}}^2 + \Delta\mu_{\text{true}}^2}{2\sigma_{\Delta\mu}^2}\right] I_0\left(\frac{\Delta\mu_{\text{obs}} \Delta\mu_{\text{true}}}{\sigma_{\Delta\mu}^2}\right), \end{aligned} \quad (34)$$

the distribution for u is

$$\begin{aligned} \text{Rice}(u_j | u_{\text{true}}, \sigma_{u,j}) &= \frac{u_j}{\sigma_{u,j}^2} \\ &\times \exp\left[-\frac{u_j^2 + u_{\text{true}}^2}{2\sigma_{u,j}^2}\right] I_0\left(\frac{u_j u_{\text{true}}}{\sigma_{u,j}^2}\right), \end{aligned} \quad (35)$$

where I_0 is the zeroth-order modified Bessel function. Integrating over $\tilde{u}$ yields

$$\begin{aligned} p(u_j | m_{\text{tot}}) & \\ &= \frac{1}{\sqrt{m_{\text{tot}}/M_{\odot}}} \int_0^\infty \text{Rice}\left(\frac{u_j}{\sqrt{m_{\text{tot}}/M_{\odot}}} \middle| \tilde{u}, \frac{\sigma_{u,j}}{\sqrt{m_{\text{tot}}/M_{\odot}}}\right) p(\tilde{u}) d\tilde{u}. \end{aligned} \quad (36)$$

To allow for outliers we adopt a mixture model for each datum,

$$\begin{aligned} \mathcal{L}_j^{\text{mix}} &= \int \left[(1 - f_{\text{outlier}}) \cdot \frac{1}{\sqrt{m_{\text{tot}}/M_{\odot}}} \cdot p(\tilde{u}) \right. \\ &\quad \left. + f_{\text{outlier}} \cdot \frac{1}{\sqrt{m_{\text{tot}}/M_{\odot}}} \cdot \frac{1}{C_o} \cdot G_o\left(\tilde{u} | \mu_o, \sigma_o^2\right) \right] \\ &\quad \times \text{Rice}\left(\frac{u_{\text{obs}}}{\sqrt{m_{\text{tot}}/M_{\odot}}} | \tilde{u}, \frac{\sigma_{u,j}}{\sqrt{m_{\text{tot}}/M_{\odot}}}\right) d\tilde{u}. \end{aligned} \quad (37)$$

The mass-luminosity relation can thus be constrained by substituting m_{tot} with

$$m_{\text{tot}} = \mathcal{M}^{-1}(M_{\text{G},1}, Z; \Theta) + \mathcal{M}^{-1}(M_{\text{G},2}, Z; \Theta) \quad (38)$$

5.2 Metallicity marginalization

At metallicity node Z_q , the component masses, total mass, and dynamical mass scale are

$$m_{jkq} = \mathcal{M}^{-1}(M_{G,jk}, Z_q; \Theta), \quad M_{\text{tot},jq} = m_{j1q} + m_{j2q}, \quad s_{jq} = \sqrt{M_{\text{tot},jq}/M_{\odot}}. \quad (39)$$

For one system, the final Stage-2 likelihood is

$$\mathcal{L}_j^{\text{dyn}}(\theta_M) = \sum_{q=1}^{81} \bar{P}_{jq} \mathcal{L}_j^{\text{mix}}[s_{jq}(\theta_M)]. \quad (40)$$

This sum propagates the system’s Stage-1b metallicity uncertainty into its mass and dynamics. It is not a second use of the spectroscopic or CMD data. Population-distribution factors and additional quadrature weights must not be inserted in [equation \(40\)](#), because both are already present in the probability masses $\bar{P}_{jq}$.

6 Parameters, priors, and posterior computation

The sampled MLR coordinates are

$$\theta_M = \left(c_0, \{a_h\}_{h=2}^{K_Z}, \{b_i\}_{i=2}^{K_x}, \{r_{ih}\}_{i,h \geq 2}, \log \lambda_x, \log \lambda_Z, f_{\text{outlier}} \right), \quad (41)$$

giving 35 scalar degrees of freedom for $K_x = 8$ and $K_Z = 4$. The priors and fixed regularization scales are summarized in [table 2](#). Here $\Pi_{11} = 0.00703$, and the PARSEC-derived reference increments are $\lambda_{x,\text{ref}} = 0.18624$ and $\lambda_{Z,\text{ref}} = 0.13078$ dex.

Table 2: Frozen Stage-1a, Stage-1b, and Stage-2 parameter specification. Normal distributions are written as location and standard deviation.

Stage	Quantity	Prior or fixed value	Role
1a	θ_{rel}	MLE; no prior	Difference calibration (b_G, s_z, ν); fixed in Stage 1b
1b	δ_z	$\mathcal{N}(0, 0.3)$ dex	JCAPS–PARSEC zero point
1b	s_C	Half-Normal(0.1) mag	CMD excess colour scatter
1b	ω	Dirichlet(1, ..., 1)	Six-component population weights
2	c_0	$\mathcal{N}(\Pi_{11}, 0.10)$ dex	First total-surface coefficient
2	a_h, b_i, r_{ih}	$\mathcal{N}(0, 1)$	Unconstrained monotone coordinates
2	$\log \lambda_x$	$\mathcal{N}(\log 0.18624, 0.7)$	Magnitude increment scale
2	$\log \lambda_Z$	$\mathcal{N}(\log 0.13078, 0.7)$	Metallicity increment scale
2	f_{outlier}	Beta(3, 12)	Effective component weight

Stage 1b and Stage 2 were sampled with the No-U-Turn Sampler (NUTS) ([Hoffman and Gelman, 2014](#)), using four chains and 1,000 retained draws per chain. The frozen fits had no divergent transitions. Posterior adequacy must additionally be assessed with rank-based $\hat{R}$, effective sample sizes, chain traces, and posterior predictive or held-out checks; successful execution alone is not scientific validation.

7 Information propagated, assumptions, and limitations

[Table 3](#) distinguishes uncertainty that is represented in the current likelihood from uncertainty held fixed or omitted. In particular, the model propagates the reported per-star JCAPS errors, heavy-tailed metallicity scatter, CMD colour scatter, the single-system metallicity posterior, and the reported dynamical amplitude error. It does not propagate uncertainty in the absolute magnitudes, extinction correction, PARSEC model family, age distribution, or uncertainty in the preliminary Stage-1a relative-calibration estimates.

Table 3: Uncertainty and information flow in the frozen model.

Category	Represented	Not represented or deliberately cut
Metallicity	Raw per-star errors, shared Z_j , Student- t tails, CMD excess scatter, population weights, and each system's $\bar{P}_{jq}$	Uncertainty in the fixed preliminary Stage-1a estimates b_G , s_z , and ν ; cross-system covariance induced by shared Stage-1b parameters
Photometry and models	Formal BP/RP colour errors and fitted s_C	M_G uncertainty, extinction uncertainty, age as a latent variable, PARSEC-family/model discrepancy
Dynamics	Rice convolution with reported $\sigma_{u,j}$ and latent velocity component distributions	Uncertainty in the fixed good/outlier shapes and additional astrometric covariance beyond the reported $\sigma_{u,j}$
Feedback	Metallicity uncertainty is passed from Stage 1b to Stage 2	Dynamics cannot update δ_z , s_C , ω , or individual metallicity posteriors

Several interpretation boundaries follow. First, the shared Z_j is a model assumption for a co-natal binary, not direct proof that both stars have identical surface abundances. Second, the 4000-K threshold expands CMD coverage but may also increase sensitivity to low-temperature PARSEC colour systematics. Third, hard monotonicity encodes the adopted physical ordering of the MLR; it does not demonstrate that PARSEC or the observations are unbiased over the full domain. Fourth, the cut reduces feedback from a potentially misspecified dynamical model, but it understates joint uncertainty by fixing the Stage-1a estimates, discarding the shared Stage-1b posterior covariance, and excluding any information that the dynamics could provide about metallicity calibration. A fully joint analysis would be a different model, not merely a more accurate implementation of the present one.

A Constructive monotone reparameterization

The total coefficient grid Θ is generated from unconstrained raw variables rather than by rejecting proposals that violate inequalities. Define $\text{softplus}(t) = \log(1 + e^t)$ and $\text{sigmoid}(t) = (1 + e^{-t})^{-1}$. With positive scales $\lambda_x = e^{\log \lambda_x}$ and $\lambda_Z = e^{\log \lambda_Z}$, construct the first row and column as

$$\Theta_{11} = c_0, \quad (42)$$

$$\Theta_{1h} = \Theta_{1,h-1} + \lambda_Z \text{softplus}(a_h), \quad h = 2, \dots, K_Z, \quad (43)$$

$$\Theta_{i1} = \Theta_{i-1,1} - \lambda_x \text{softplus}(b_i), \quad i = 2, \dots, K_x. \quad (44)$$

The first row therefore increases with Z , whereas the first column decreases with x . Interior points are visited in increasing $i + h$. For each point, define

$$L_{ih} = \Theta_{i,h-1}, \quad U_{ih} = \Theta_{i-1,h}, \quad (45)$$

and set

$$\Theta_{ih} = L_{ih} + (U_{ih} - L_{ih}) \text{sigmoid}(r_{ih}), \quad i, h \geq 2. \quad (46)$$

If all previously constructed points obey the two monotone orders, then

$$L_{ih} = \Theta_{i,h-1} \leq \Theta_{i-1,h-1} \leq \Theta_{i-1,h} = U_{ih}. \quad (47)$$

Thus [equation \(46\)](#) lies inside a non-empty interval and simultaneously satisfies $\Theta_{i,h-1} \leq \Theta_{ih} \leq \Theta_{i-1,h}$. Induction over $i + h$ therefore proves the coefficient inequalities in [equation \(24\)](#) for the complete grid. Finally, $D = \Theta - \Pi$ is computed deterministically.

For completeness, the derivative identity for a degree- d_x B-spline surface has the form

$$\frac{\partial g}{\partial x} = \sum_{i=1}^{K_x-1} \sum_{h=1}^{K_Z} \rho_{x,i} (\Theta_{i+1,h} - \Theta_{ih}) \tilde{B}_i(x) C_h(Z), \quad \rho_{x,i} > 0, \quad (48)$$

where $\tilde{B}_i$ is a non-negative degree- $(d_x - 1)$ basis function. The corresponding Z derivative contains $\rho_{Z,h} (\Theta_{i,h+1} - \Theta_{ih}) B_i \tilde{C}_h$ with $\rho_{Z,h} > 0$. The coefficient orders therefore establish the derivative signs everywhere in the spline domain, including the appropriate one-sided limits at its boundaries.

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